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Traffic Accident Diagnosis in the last decade - A case study

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***Abstract*— Accident reduction projects must undergo the research stage of the current situation and trends based on a historical record of accidents. Identification of hot spots for accidents is a vital step in safety management, since roadway improvements are supposed to be applied to random locations. In this context we present a diagnosis of the traffic accidents in the Cidade Industrial de Curitiba, CIC (Brazil), and to identify several measures that could contribute to the reduction of accidents. We take advantage of GIS and exploratory analysis to identify road accident hot spots both visually and statistically. Finally, some guidelines are suggested, using 10 years of data in a case study.**

I. Introduction

In the construction of a sustainable city, issues like urban mobility must be featured during the entire urban planning process. Road accidents are considered one of the most negative impacts of developing modern transportation system, resulting in loss of lives.

Road traffic injuries are the eighth leading cause of death globally, and the leading cause of death among young people aged 15-29 [1]. Current trends suggest that by 2030 road traffic deaths will become the fifth leading cause of death unless urgent action is taken [1], [2]. In 2012, approximately 45 thousand people in Brazil died in traffic accidents [3], and around 16.1 billion reais were spent due to accidents, with 10.7 billion Brazilian reais were spent on accidents resulting in fatalities and 5.4 billion reais spent with injured victims [1].

Traffic safety engineers face an important question: where to implement safety precautionary measures in order to impact traffic safety. "Hot Spots", "Black Spots" or "High accident locations" are sites on the section of roads and highways with higher accident frequency [4].

In this paper we address this problem by the integration of open data for urban transportation, exploring a case study in Brazil. The main goal is to discover what the data can tell us about the observed mobility plan with the help of visualization and display techniques of Geographical Information System (GIS). The rest of this document is organized as follows: Section 2 contains an overview of a motivating example. Section 3 presents the related work. Data is presented within Section 3. Behavioral trends are presented in Section 4. Conclusions and future work are presented in Section 5.

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[1] http://www.cnt.org.br/Paginas/Agencia_Noticia.aspx?noticia=brasil-gasta-16-bilhoes-porano-em-acidentes-de-transito-emds-brasilia-09042015 Last accessed on 15/05/2016

II. A Motivating Example

Consider Curitiba, a city located in the south of Brazil, housing 1.8 million people in a total area of 430,9 km^2, according to the Brazilian Institute of Geography and Statistics (IBGE) [2]. This area encompasses 75 districts, and is surrounded by other 29 municipalities, known as the Metropolitan Region of Curitiba (with the Portuguese acronym *RMC*).

The so-called Industrial City of Curitiba (in portuguese, the acronym CIC) is the largest of the 75 districts of Curitiba. As the name suggests, CIC is an industrial district, but also accommodates other uses including residential. It is strategically located in the western portion of the city, between two national highways (166 and 277 - Figure 1). CIC occupies 43.7 km^2, representing approximately 10% of the total Municipal area. Its area is distributed as following: 21 km^2 for Industry, 6.7km^2 for residential areas, 3.86 km for road systems, 5.14 km^2 for trade areas, 1.4 km^2 for service and 1.6 km^2 for residential and service areas [5].

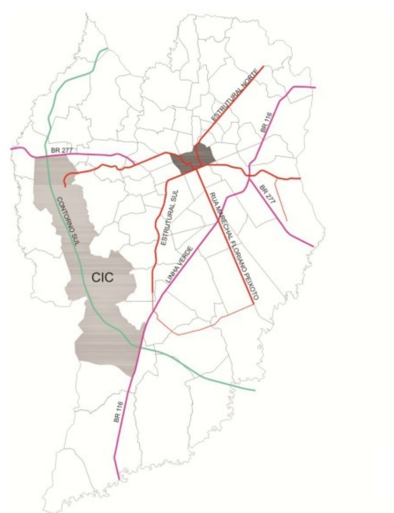

Fig. 1: Location of CIC and downtown area.

Curitiba created tax incentives to encourage the development of CIC, offering, for example, taxes exemption (such as Brazilian taxes ICMS and IPTU), with direct and long-term financing. Any company, when deciding to settle at CIC, would choose a plot, and the Local Government would offer area demarcation, earthwork and infrastructure services [3]. After the oil crisis in 1979, the lack of understanding of

[2] http://www.ibge.gov.br – Last visited on 14/05/2015.

[3] http://www.agenciacuritiba.com.br/publico/conteudo.aspx?codigo=13 Last accessed on 16/05/2016.

State and Municipal Governments and lack of guidance from the National Government resulted in uncertain territory policies in the district for almost two decades [6]. With large vacant or underused areas, many land invasions and slums constructions occurred.

Considering the information listed above, several questions might arise, such as, which areas concentrate the traffic injuries, how to explore the main hotspots to plan smartest precautionary measures, how the hotspots are distributed around the neighborhood and along the decades, among others.

Briefly, the efforts of integrating different domains (data, urban planning, GIS) may vary in several directions: (i) from the GIS perspective, how to explore algorithms, statistics and visualization with efficiency; (ii) from the integration perspective, the challenge is standardization and not to have a bottleneck; (iii) from the city perspective, the challenge is how to learn from the past (using different techniques) in order to have a smarter city.

III. Related Work

Each year, around 1.9 million people die in road accidents; between 20 to 50 million are injured, many of whom are left with lifelong disabilities [4]. Taking this into consideration , the United Nations launched the Decade of Action for Road Safety (2011-2020) [5] with the objective of reducing the number of fatalities and injuries. To achieve this goal, member countries launched the Global Plan for the Decade of Action for Road Safety, which has five pillars of action: (1) building road safety management capacity; (2) improving the safety of road infrastructure and broader transport networks; (3) further developing the safety of vehicles; (4) enhancing the behavior of road users; and (5) improving post-crash care.

High-income countries can improve their performance and share experiences. For example, in the decade of 1990, Netherlands developed a program called Sustainable Road Safety [7], where the highway system should be as easy as possible to understand and use. Thus, when users make mistakes, the chance of this result in accidents is the lowest possible (called pardonable road system).

Regarding data, there are some sites available (such as crash map [6], or the UK Department of Transport [7]) related to road safety, but not all of them are available for download (bringing additional challenges to research at this area).

Contributing Factors to Traffic Accidents. According to [8], traffic accidents occur due to the interaction of adverse events present in the moment of the accident and can be divided into three groups: i) human factor; ii) road factor/environment; and iii) vehicle factor. Other factors might impact [8], indicating greater risk during late evening-early morning hours every day of the week and the greatest risk during Friday–Saturday and Saturday–Sunday evening-to-morning hours.

Accident Accumulation Zones. Usually called black spots or hot spots, the Accident Accumulation Zones (AAZ) are locations with a high potential for road accidents. Each country uses the most compatible criteria with the local reality to identify AAZ [9], as show in Table III. No method is free of limitations; each chosen method must consider the data, financial, and human resources available. In order to minimize flaws, the use of more than one method is advised.

Country	Section length	Frequency
Austrália	Fairly short	At least 3 casualty crashes in 5 years
England	300 meters	12 crashes in 3 years
Germany	300 meters	8 crashes in 3 years
Norway	100 meters	4 crashes in 3 years
Portugal	200 meters	5 crashes in 3 years
Thailand (DOH)	Vary	At least 3 crashes per year

TABLE I: Accident Accumulation Zones by Country [9].

A. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a philosophical approach to data analysis [10]. The posed question is: what data can tell us about certain relationships, properties or structures. There are no imposed techniques to apply to the data set, but graphical visualization plays an important role in this approach [11]. The non-inferential approach in data analysis encourages the openness perspective required to integrate new domains. The availability of processing power and data storage provide new tools for handling massive amounts of data processing, allowing flexible search of evidences in the available data through designed experiments [12].

B. Geographical Information System - GIS

GIS stores data from heterogeneous sources in several formats in the form of geodatabases representing spatial features, storing latitude and longitudinal positions. An enterprise geodatabase can help solve common transportation challenges: the many origins, destinations, paths, and conveyances that may be present, along with their locations.

In particular, geographic information systems for transportation (GIS-T) are interconnected hardware, software, data, people, organizations and institutional arrangements for collecting, storing, analyzing, and communicating particular types of information about the earth [13].

Within transportation systems, the GIS database for transportation is organized on the basis of the road network map. Each link can have as attributes topographic (e.g. nodes UTM coordinates, total length), toponomastic (street names),

[4] http://www.who.int/roadsafety/decade_of_action/plan/plan_spanish.pdf Last accessed on 18/05/2016.

[5] http://www.who.int/roadsafety/decade_of_action/en/ Last accessed on 18/05/2016.

[6] http://www.crashmap.co.uk/ Last accessed on 18/05/2016.

[7] https://data.gov.uk/dataset/road-accidents-safety-data Last accessed on 18/05/2016.

[8] http://www.rita.dot.gov/bts/sites/rita.dot.gov.bts/files/subject_areas/safety/vehicle_fatality Last accessed on 18/05/2016.

physical (traffic directions, number of lanes), transport (road typology by means of the speed-flow curves) and transit (description of public transport lines and corresponding frequencies) information [14]. Here, generally origin destination matrices express the traffic demand.

IV. Methodology

The methodology used for the exploratory data analysis comprised the following steps: (i) the acquisition/characterization of data sources; (ii) the data integration and processing; (iii) the data analysis (along with hypothesis and limitations); and (iv) the definition of parameters/challenges which impact the proposed problem solution.

A. Data Sources

Two datasets where used: (i) Source 1: Integrated Service Care for Trauma Emergency (with the portuguese acronym SIATE [9]) and (ii) Source 2: Institute of Research and Urban Planning of Curitiba (with the portuguese acronym IPPUC [10]).

The SIATE data ranged yearly from 2005 to 2014, and contained the following items: i) Transportation modes involved and type of collision; ii) Date and time of the collision; iii) Place of occurrence (district, road or reference point); iv) Gender (male or female); v) Identification of victim (passenger, pedestrian, driver, backseat, etc); and vi) Accident code (code 1- accident with light injuries; code 2 – accident with severe injury without life risk; code 3 – accident with severe injury with life risk; code 4 – deceased).

From IPPUC, the data was from streets, bus stops and health facilities (hospitals and health care units) from the city. The collected information focuses only on the place and during the route to the hospital, not including information of whether the victims from accidents with severe injury died later at the hospital.

B. Data Integration and Processing

The complete IPPUC dataset along with source 1 (extracted from excel files) were inserted in a PostGIS [11] database. Different sources were created as different tables in the database. Later, specific tablespaces and indexes were created in order to optimize the access. Since semantic errors were present (different sources presented different street names for the same location, for example [15]), geolocation and specified time range were used to correlate the data. After the information was received, the data was grouped to generate accident profiles.

C. Data Analysis

The objective was to analyze the district of CIC, and using the available datasets, locate the hotspots. The data analysis process used the QGIS [12] and ArcGIS[13].

[9] `http://www.bombeiros.pr.gov.br/modules/conteudo/conteudo.php?conteudo=151` Last accessed on 18/05/2016.

[10] http://www.ippuc.org.br/ Last visited on 14/05/2015.

[11] http://www.postgis.net Last visited on 15/05/2014.

[12] http://www.qgis.org Last visited on 15/05/2015.

[13] https://www.arcgis.com/ Last visited on 15/05/2015.

Curitiba has 9.135 streets, divided among 17 categories (Anel Central, Central, Coletora 1, Coletora 2, Coletora 3, Externa, Linhão, Normal, Outras Vias, Pedestre, Prioritária 1, Prioritária 2, Rodovia Estadual Duplicada, Rodovia Estadual Simples, Rodovia Federal Duplicada, Rodovia Federal Simples, and Setorial). CIC is the district which concentrates the majority of them, with 1217 different ones.

The city has 68 private hospitals, and 7 hospitals from the government. Within the CIC district, only two private hospitals are available (one was already closed by this article submission, and the other has a small size category), along with a third particular (neurology specialist), available at a nearby district (Campo Comprido). The city also has 126 health care units (16 at CIC), but they are not prepared to receive injured people from accidents.

Within 2014, 10% of the accidents of Curitiba were in CIC. Historically, it is the district with higher numbers of accidents. Figure 2 present these numbers, from 2005 till 2014.

Looking back at the same years, the top ten types of accidents include several categories, as shown in Figure 3. Among them, the top 3 categories include vehicles, motorcycles and tramplings. Considering all data range, the top type of accident is still motorcycle vehicle collision. Among them, the top 4 categories include vehicles, motorcycles and tramplings. Considering all data range, the top type of accident is still motorcycle vehicle collision.

Still considering the same data range, we have the injuries by gender (Figure 4) and type of injuries (Figure 5). Statistics shows that men have been affected almost three times more than women along the region in the last the decade. Nevertheless, the majority of the injuries include accident with severe injury without life risk, and the cases with unharmed victims are almost non existent. Along the years, the injuries have been not equally divided concerning the time of the day when they mostly occurred, as shown in Figure 6.

Next, we divided CIC district in 1000 by 1000 meters quarters to better identify the Accumulation Zones. The information used was from accidents that occurred in 2012, 2013, and 2014, from SIATE.

If we consider only top three type of injuries within 2012-2014, for SIATE, some regions keep presenting the highest number of injuries, as showed in Figure 7, in yellow. The regions highlighted within the district presented the worst concentration of injuries.

As shown in Figure 7, three regions had more than 15 accidents for three consecutive years (2012, 2013, 2014). Within these regions is located Street Eduardo Sprada, Street Rua Raul Pompéia and Street Senador Accioly Filho, which cross street Juscelino Kubitscheck de Oliveira.

*1) **Avenue Juscelino Kubitscheck de Oliveira and Street Eduardo Sprada.**:* Figure 8 presents one of the AAZ, located at Avenue Juscelino Kubitscheck de Oliveira and Street Eduardo Sprada. Note that the location has Volvo, Havan, MC Donald's, a gas bus station within the region, and keeps concentrating accidents for the last three years. The same figure presents the locations with higher number of

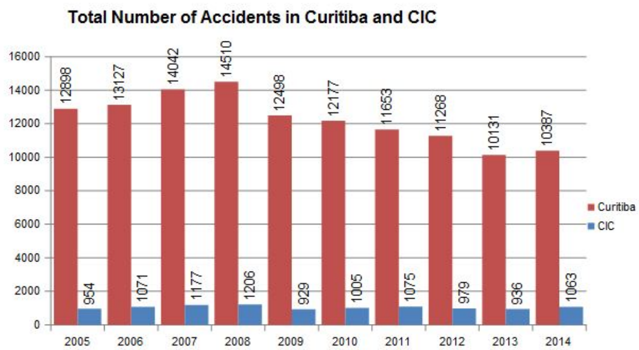

Fig. 2: Total number of accidents, from 2005 till 2014 in Curitiba and CIC.

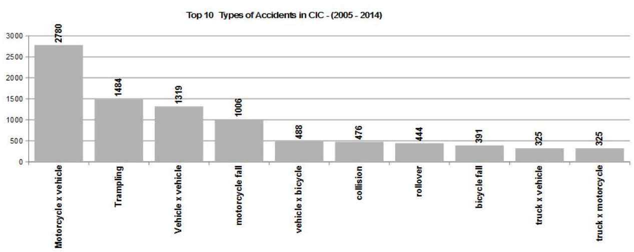

Fig. 3: Top 10 types of accidents in CIC.

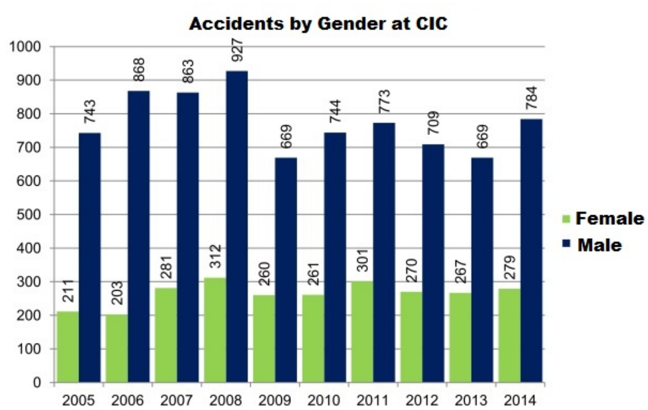

Fig. 4: Injuries by gender in CIC.

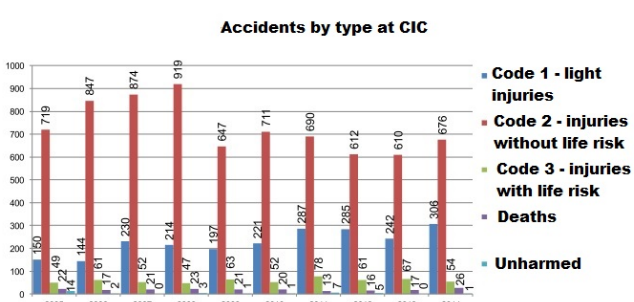

Fig. 5: Injuries by type in CIC.

injuries, along with the locations of hospitals and health care units. Figure 8 presents other important issue: the post-cash

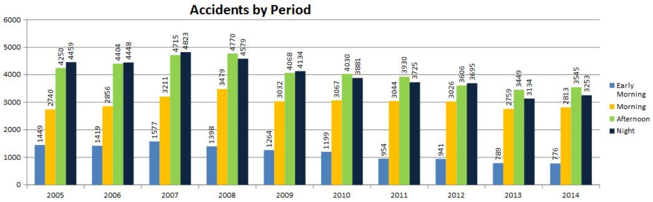

Fig. 6: Accidents by period in CIC.

care is impacted, since from the three hospitals located at the area, one was already closed, the other manages neurology specialties, and the third one is a small size hospital.

Several items were present within this crossing, such as:

- lack of road maintenance (such as areas with too much water in Figure 9-C);
- bad sign conditions, and in particular, bad signs for accessing establishments (as shown in Figure 9-B);
- regions for pedestrians in poor condition (as shown in Figure 9-D);
- lack of traffic lights for pedestrians (Figure 9-C);
- damaged/lack of vertical signs (Figure 9-A);
- location of bus stops/routes without infrastructure for pedestrians; and
- disrespect to transit laws(as shown in Figure 9-D).

V. Results and Discussion

As a result from the traffic accidents profile, the following results were acquired: to the data from SIATE, the main collisions were between automobiles and motorcycles. The driver and male gender were the main participants, and the accidents considered serious without risk to the victim were the most frequent.

The majority of accidents happened at night and the highest time peak was from 18h to 19h59min. The established AAZ were at the following intersections: Ruas Eduardo Sprada, Raul Pompéia and Rua Senador Accioly Filho with Av. Juscelino Kubitscheck de Oliveira.

Several considerations might be suggested in order to reduce traffic accidents (not only in CIC, but other hot spots at the city [16]): 1. Build a trustable database of injuries; 2. Implement the same methodology of approach by the responsible entities for collecting data, so homogeneous information can be generated; 3. Include road signs where they are missing, and build and improve the infrastructure available to pedestrians; 4. Provide regular inspections so the regulations are followed; and 5. Promote population awareness.

In order to provide better reports at determined places like AAZ, further information is necessary, such as data on: 1. Kind of collision (rear-end, side impact, etc); 2. Contributing Factors, Medium Daily Traffic Volume; and 3. Estimates of gravity index.

From the computer science perspective, some of the detected challenges are as follows: (i) Data Integration: different coordinate systems and formats are available. The majority do not provide metadata, and some of the sources

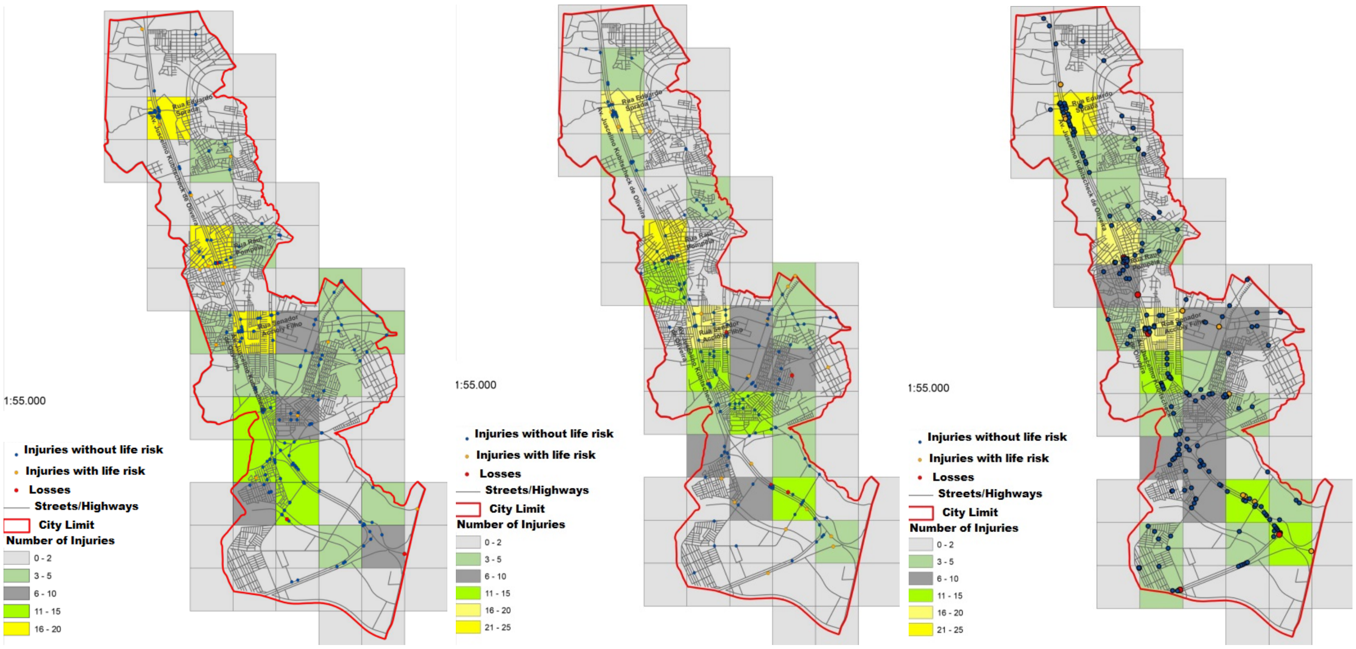

Fig. 7: Accidents at CIC: 2012(left), 2013(center), 2014(right).

Fig. 8: The heatmap of accidents in CIC, with zoom at the first black spot.

might be outdated (such as hospitals which are already closed); (ii) Knowledge domain and in loco verification: only textual and spatial information were not enough to understand the reasons for the accident locations (shown in Figure 9); (iii) Lack of Data Sources: data such as provenance of accident locations to hospitals is still no public

Fig. 9: Issues at the first black spot.

available.

Beyond what has been pointed out, a more in-depth study of the surrounding area should be implemented, examining land use, infrastructure and human behavior.

VI. Conclusion

Research in traffic accident diagnosis and is not recent, but the exploration through different domains is still an ongoing effort. The possibility of implementing models within GIS and integrating them with different sources provides planners with a powerful and flexible tool for analyzing the city along with their highways, and deciding on smartest ways of using resources.

This paper presents the concepts, applications, and challenges of analyzing traffic accidents. Later these definitions are explored in a practical case study, within the Curitiba metropolitan region, Brazil, with data analysis aimed at supporting planners.

Future work includes the integration of additional data, and the use of personalization and recommendation techniques in order to explore the data.

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